Joint-Embedding Predictive Architectures and Latent Residual Conditioning for Multivariate Time-Series Anomaly Detection

Andre Williams, Imadeldin Mahgoub, *Senior Member, IEEE,* and Hamid Akbarian

***Abstract*—Deep reconstructive detectors for multivariate time-series anomaly detection minimize observation-space error, fitting stochastic sensor noise that produces heavy-tailed residuals and severe false-alarm flooding under automated threshold calibration. We investigate how Joint-Embedding Predictive Architectures (JEPAs) condition predictive residuals in a low-dimensional latent space under distribution shift, alongside an Operator-Entropy variant (OpEntropy JEPA) combining a linear Koopman transition operator, spectral entropy and shifted logarithmic Bregman barrier regularizers, coordinate saliency gating, and a multiscale dyadic frequency front-end. Across 33 continuous telemetry streams from seven physical datasets under chronological held-out Streaming Peaks-Over-Threshold calibration, our primary finding is relative false-alarm suppression rather than universal detection superiority or exact attainment of the 0.1% nominal design risk. Latent predictive scoring reduces held-out false-positive rates three- to four-fold under retrospective buffered attribution (7.4% to 10.1% versus 30.0% to 30.7% for TimesNet and TranAD) and eight- to twenty-fold under strictly causal streaming (1.1% to 2.8% versus 22.2% to 24.3%), while point F1 and precision-recall ranking remain comparable to reconstructive baselines. Controlled ablations isolate two distinct mechanisms: LayerNorm bottlenecks compress residual kurtosis, whereas learned latent projections halve calibration-to-holdout distribution drift relative to normalized observation forecasters. On a fixed Koopman backbone, equivalence testing confirms that spectral entropy and barrier terms preserve full operator rank without altering point F1 within a 0.01 to 0.03 margin, while the OpEntropy transition architecture shortens detection delay relative to unconstrained TS-JEPA. Static thresholding still overshoots its 0.1% design target by 15- to 64-fold out of sample, driven primarily by inter-block distribution shift and serial dependence.**

***Index Terms*—Time-series anomaly detection, Multivariate analysis, Threshold calibration, Extreme value theory, Representation learning.**

I. Introduction

Autonomous anomaly detection in multi-sensor telemetry monitors operational health across aerospace systems, Earth observation, cyber-physical infrastructure, and wearable digital healthcare [1], [2]. In complex deployments such as NASA SMAP and MSL rovers, industrial process loops, and biometric sensor arrays, dozens of channels concurrently track power buses, thermal gradients, vibrations, and kinematics [3]. Operational monitors must identify subsystem faults without overwhelming operators with nuisance alerts.

False-Alarm Flooding Under Automated Threshold Calibration: Operational monitoring requires automated decision rules that convert continuous anomaly scores into binary alarms using nominal calibration data alone. Prevailing deep detectors (TimesNet, TranAD, Anomaly Transformer) minimize point-level reconstruction error in raw observation space [4]–[6]. Because raw telemetry contains ambient sensor jitter and channel-scale disparities, observation-space residuals exhibit heavy-tailed excess kurtosis ($\gamma_2 > 8$) and strong serial clustering. When paired with automated extreme-value thresholding such as Streaming Peaks-Over-Threshold (SPOT) [7] at nominal design risk $q = 10^{-3}$ (0.1%), these heavy-tailed residuals suffer severe calibration breakdown on unseen nominal data, triggering false alarms across 22%–31% of nominal timesteps.

Latent Residual Conditioning and Scope of Claims: To suppress false-alarm flooding without synthetic anomaly injection [8], we study how Joint-Embedding Predictive Architectures (JEPAs) condition predictive residuals in a low-dimensional latent space [9] under distribution shift, and formulate an *Operator-Entropy JEPA* (OpEntropy JEPA) transition architecture regularized by Koopman spectral entropy and a shifted Bregman energy barrier. Rather than claiming that physics-inspired regularizers drive anomaly detection accuracy, we evaluate them strictly as algebraic conditioning priors on abstract latent coordinates. Our evaluation separates three operational questions: (i) relative false-alarm suppression under automated calibration versus exact attainment of the nominal 0.1% risk target, (ii) discriminative detection across operating points and fault classes, and (iii) structural score-emission latency versus empirical fault-detection delay.

A. Contributions

This study makes four focused contributions:

1) **Primary Mechanism Isolation (Compression vs. Drift):** Across 12 controlled backbone arms on all 19 non-GHL streams (Section VI-E, Fig. 3(a), Table I(e)), adding a D=32 LayerNorm bottleneck to an observation forecaster compresses calibration kurtosis (γ_2 : $6.10 \rightarrow 0.91$) and halves causal false-positive rates ($24.7\% \rightarrow 11.7\%$), while

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learned latent projection in TS-JEPA and OpEntropy JEPA halves calibration-to-holdout Kolmogorov–Smirnov drift (D_{KS} : $0.376 \rightarrow 0.209$–0.214) and lowers causal false alarms to 0.9%–2.8%.

2) **Relative False-Alarm Suppression vs. Detection Limits:** Across 33 streams from seven datasets (Table II), latent scoring reduces relative held-out false-alarm rates three- to four-fold under buffered attribution (7.4%–10.1% vs. 30.0%–30.7%) and $8\times$–$20\times$ under strictly causal streaming (1.1%–2.8% vs. 22.2%–24.3%). Separating deployable calibration quantiles (q^*) from post-hoc matched realized test-FPR curves (f^*, Fig. 3(b)) shows higher low-FPR causal event recall (0.395–0.431 vs. 0.217 at $f^*{=}0.01$), whereas Point-F1, PR-AUC, and buffered recall match reconstructors and quiet flatlines remain weakly detected (0.02–0.07 recall).
3) **Operator-Entropy Conditioning and Equivalence Bounds:** Proposition 1 establishes non-triviality conditions for the Koopman transition operator under finite-ϵ and singular energy barriers. While the OpEntropy transition architecture cuts buffered detection delay from 110.3 to 79.5 steps ($127.0 \rightarrow 58.2$ causal), fixed-backbone Two One-Sided Tests ($\lambda_{\text{ent}}{=}0$ control, Table I(a)) confirm Point-F1 equivalence within $\Delta = \pm 0.03$ and down to ± 0.010, isolating the spectral regularizer as a conditioning constraint rather than a detection driver.
4) **Operational Calibration Gap and Refitted Permutation Audit:** Static SPOT misses its nominal 0.1% target out of sample (1.51%–1.95% stream-macro causal and 5.54%–6.38% buffered FPR across 33 streams for JEPAs, or 15–$20\times$ and 55–$64\times$ overshoot vs. 142–$222\times$ for reconstructors, Table IV). A three-stage refitted permutation decomposition (Table IV(c)) estimates that inter-block shift accounts for 56.6%–64.1% of excess causal overshoot for JEPAs (77.5%–92.3% for reconstructors) and serial dependence for 29.1%–35.3%.

II. Related Work

Dynamical Systems, Reconstruction, and Forecasting Detectors: Multi-sensor telemetry models system dynamics as state trajectories $\dot{x}(t) = \mathbf{f}(x(t))$ [10]. Neural ODEs [11] parameterize velocity fields in raw observation space, where integrating through high-frequency sensor noise induces numerical stiffness. Koopman theory and dynamic mode decomposition [12]–[14] learn linear operators on observable embeddings, identifying disruptions via spectral decoherence; our proposed Operator-Entropy JEPA architecture transfers this principle to a latent transition operator regularized by spectral entropy.

Reconstructive detectors model nominal patterns via observation reconstruction: LSTM forecasting with dynamic thresholding for spacecraft telemetry [3], association-discrepancy attention in Anomaly Transformer [4], 2D FFT-periodicity variation in TimesNet [5], and adversarial sub-series attention in TranAD [6]. These share a vulnerability: point-level losses bias models toward stochastic noise, accumulating residual variance across irrelevant channels in high-dimensional arrays.
Threshold Calibration, Contrastive, and Self-Supervised Detection: Unsupervised monitoring requires automated rules to map residual scores to binary alarms without labels. Non-parametric thresholding [3] and Streaming Peaks-Over-Threshold (SPOT) [7] fit Generalized Pareto tails under design risk q; however, EVT asymptotic guarantees assume i.i.d. samples, which temporal autocorrelation and operational drift violate.

To avoid reconstructing noise, contrastive methods train discriminative representations: NCAD [8] injects synthetic anomalies, yielding lower calibrated point detection than predictive models (0.058 vs. 0.160 macro Point-F1 across 11 datasets, Table II(d)). DCdetector [15] maximizes dual-attention consistency (0.070 macro Point-F1). Joint-Embedding Predictive Architectures [9], [16], [17] predict target representations in latent space with non-contrastive collapse prevention [18]–[20]. Time-series extensions [21] evaluate ranking or post-hoc threshold sweeps without studying how latent residuals behave under automated tail calibration.
Physics-Inspired Priors and Conceptual Positioning: PINNs [22] and Hamiltonian/Lagrangian networks [23], [24] embed physical conservation laws as rigid differential constraints; their stiff competing gradients [25] and known-PDE requirements fit poorly to uncalibrated multi-sensor telemetry. We strictly distinguish *physics-inspired* algebraic regularizers (where latent coordinates are abstract learned features, not physical state variables) from *physics-informed* conservation constraints; *causal data access* from *score-emission latency*; and *operator non-degeneracy* from *encoder representation diversity* and *anomaly separability*.

III. Multi-Sensor IoT and Remote Sensing Benchmark Suite

Our benchmark suite comprises 11 datasets spanning 45 continuous telemetry streams sourced from NASA spacecraft archives [3], the UCI Machine Learning Repository [26], [27], TimeEval [28], and mTSBench [29]: 33 streams are evaluated under the primary multi-seed held-out calibration protocol and 12 reserved for operational case studies. Streams span four physical domains: *spacecraft telemetry* (SMAP and MSL entities, with two short-nominal MSL streams reserved; SWAN-SF solar magnetograms [30]); *wearable kinematics* (six Daphnet FoG streams [26]: 64 Hz tri-axial accelerometry for Freezing-of-Gait events, so $S{=}64$ is a 1-s horizon); *industrial processes* (14 GHL gasoil heating-loop streams, the Genesis pick-and-place testbed, and GECCO water-distribution chemistry/hydrology at 1-minute resolution); and *smart infrastructure* (Room-Occupancy environmental sensing [31], six CICIDS edge network-flow streams, CalIt2 people-flow [27], and Metro I-94 highway traffic).

Each stream provides nominal training sequences and labeled anomaly test sequences; channels are standardized by training statistics with $\boldsymbol{\sigma}$ floored at 10^{-2} against flatline phases. We evaluate the standardized subsets studied in the literature [3], [28] (of 55 SMAP and 27 MSL archive entities) for strict matching and reproducible runtime.

To eliminate circular in-sample threshold tuning, each nominal sequence is split chronologically into training (65%), nominal calibration (17.5%), and held-out nominal evaluation

(17.5%) blocks separated by an unobserved 64-step isolation gap (Section V-A). Stream inclusion in the primary multi-seed protocol required: (i) $N_{\text{cal}} \geq 500$ calibration windows after splitting, and (ii) tractable stride-1 multi-seed runtime. Exactly 33 streams across seven datasets satisfy both criteria (6 Daphnet, K=9; 14 GHL, K=16; 4 MSL, K=55; 6 SMAP, K=25; Genesis, K=18; Metro, K=5; SWAN, K=38). The remaining 12 streams (CalIt2, K=2; two Room-Occupancy, K=5; two short-nominal MSL; GECCO, K=9; six 72-channel CICIDS flows) appear in operational incident case studies and a secondary 11-dataset, 130-stream single-seed baseline cross-check (Table II(d)); the dyadic front-end ablation (Table I(b)) includes SMD `machine-1-2` (K=38).

Two external-validity caveats bound interpretation: (i) SMAP and MSL contain documented mislabeled segments and point anomalies [32], so per-dataset scores represent upper bounds on operational detectability; and (ii) because streams within SMAP, MSL, Daphnet, and GHL share hardware and environmental characteristics, streams within a dataset are not independent domains (N=7 dataset clusters). We therefore report unweighted Dataset-Macro means, between-stream and within-stream seed SDs, leave-one-dataset-out (LODO) sensitivity, and dataset-clustered permutation tests (Table II(c)), restricting conclusions strictly to the evaluated entities and sampling regimes.

IV. Methodology: Operator-Entropy Joint-Embedding Predictive Architecture

A. Problem Formulation and Latent Space Prediction

Consider a multivariate telemetry stream $\mathbf{X} \in \mathbb{R}^{T\times K}$ partitioned at each timestep t into a historical context window $\mathbf{x}_{\text{ctx}} = \mathbf{X}[t-C:t,:] \in \mathbb{R}^{C\times K}$ and a suspect target horizon $\mathbf{x}_{\text{tgt}} = \mathbf{X}[t:t+S,:] \in \mathbb{R}^{S\times K}$. Reconstructive detectors (TimesNet, TranAD, Anomaly Transformer) train an autoencoder to minimize pointwise observation error $\|\mathbf{X} - \mathcal{D}_\theta(\mathcal{E}_\theta(\mathbf{X}))\|_F^2$, which forces the model to reconstruct high-frequency stochastic noise across all K channels.

We instead adopt a Joint-Embedding Predictive Architecture (JEPA) [9] that shifts prediction from raw observation space into a low-dimensional latent space ($D = 32$). An online context encoder E_θ maps $\mathbf{x}_{\text{ctx}}$ to a context representation $z_{\text{ctx}} \in \mathbb{R}^D$, while a momentum target encoder E_ϕ maps $\mathbf{x}_{\text{tgt}}$ to a target representation $z_{\text{tgt}} \in \mathbb{R}^D$. The target parameters ϕ are updated via an exponential moving average (EMA) of the online parameters θ, $\phi_k \leftarrow m_k\phi_{k-1} + (1-m_k)\theta_k$ with cosine momentum schedule $m_k \in [0.996, 0.9995]$, preventing representation collapse without requiring negative contrastive pairs [16], [33]. Predicting target embeddings in latent space filters out channel-level sensor jitter and conditions the residual distribution for automated tail calibration.

B. Latent Predictor Architectures and Dynamical Inductive Priors

We formulate Operator-Entropy JEPA as our primary transition architecture and evaluate it alongside an unconstrained MLP predictor (`TS-JEPA`) and two comparative physics-inspired variants (Reynolds-Stress and Potential-Flow):

1) Operator-Entropy JEPA (Spectral Dynamics and Modal Dispersion): Motivated by Koopman operator theory [34], [35], which represents nonlinear dynamical evolution through linear operators acting on observable embeddings, we parameterize the latent transition via a learnable linear operator $\mathbf{K} \in \mathbb{R}^{D\times D}$ such that $\hat{z}_{\text{tgt}} = \mathbf{K}z_{\text{ctx}}$. To prevent $\mathbf{K}$ from collapsing onto a low-rank subspace (dimensional collapse) or shrinking toward the zero matrix (energy collapse), we regularize its singular value spectrum while decoupling modal uniformity from total operator energy.

Let $\sigma_1 \geq \cdots \geq \sigma_D \geq 0$ denote the singular values of $\mathbf{K}$, with squared modal energies $\lambda_i = \sigma_i^2$ and total Frobenius energy $\|\mathbf{K}\|_F^2 = \sum_{i=1}^D \lambda_i$. We normalize the modal energies into a smoothed probability distribution $p_i = (\lambda_i + \epsilon_{\log})/\sum_{j=1}^D(\lambda_j + \epsilon_{\log})$ with $\epsilon_{\log} = 10^{-12}$, and define the Von Neumann spectral entropy $\mathcal{H}_{\text{spectral}}(\mathbf{K}) = -\sum_{i=1}^D p_i \log p_i$. The *Spectral Entropy Deficiency* measures the Kullback–Leibler divergence of $\mathbf{p}$ from the uniform modal equipartition $\mathbf{u} = [1/D, \ldots, 1/D]^\top$:

$$\Delta\mathcal{H}(\mathbf{K}) = \log D - \mathcal{H}_{\text{spectral}}(\mathbf{K}) = \mathcal{D}_{\text{KL}}(\mathbf{p} \,\|\, \mathbf{u}) \geq 0. \quad (1)$$

Notice that $\Delta\mathcal{H}(\mathbf{K}) = 0$ if and only if all D modes carry equal energy ($\sigma_1 = \cdots = \sigma_D$), whereas rank-1 collapse yields $\Delta\mathcal{H}(\mathbf{K}) \approx \log D$. Minimizing $\Delta\mathcal{H}$ drives $\mathbf{K}$ toward an well-conditioned, near-isometric transition spectrum.

To prevent total operator energy from collapsing toward zero without introducing hard gradient clipping, we impose a smooth shifted logarithmic Bregman barrier on the normalized energy ratio $r(\mathbf{K}) = \|\mathbf{K}\|_F^2/D$:

$$\Phi_{\text{energy}}(\mathbf{K}) = r(\mathbf{K}) - 1 - \log\big(r(\mathbf{K}) + \epsilon\big) + \log(1+\epsilon) \geq 0, \quad (2)$$

with shift $\epsilon = 10^{-6}$. This barrier vanishes at the isometric scale $r(\mathbf{K}) = 1$ (where $\mathbf{K}$ is initialized near the identity) and penalizes energy decay smoothly. Combining both terms defines the spectral operator regularizer $\mathcal{R}_{\text{spectral}}(\mathbf{K}) = \Delta\mathcal{H}(\mathbf{K}) + \beta_{\text{norm}}\Phi_{\text{energy}}(\mathbf{K})$, and the complete training objective is:

$$\mathcal{L}_{\text{Op-Entropy}} = \frac{1}{BD}\sum_{b=1}^{B} \|\mathbf{K}z_{\text{ctx}}^{(b)} - z_{\text{tgt}}^{(b)}\|_2^2 + \lambda_{\text{ent}}\mathcal{R}_{\text{spectral}}(\mathbf{K}) + \beta_{\text{Lyap}}\mathcal{L}_{\text{Lyap}}, \quad (3)$$

where $\beta_{\text{norm}} = 1.0$, $\lambda_{\text{ent}} = 0.5$, and $\beta_{\text{Lyap}} = 0.02$. Here $\mathcal{L}_{\text{Lyap}}$ is a discrete Lyapunov covariance-alignment penalty that matches the trace-normalized batch covariance of predicted states $\mathbf{K}\,\text{sg}(z_{\text{ctx}})$ to that of z_{tgt}, using stop-gradient $\text{sg}(\cdot)$ on z_{ctx} so that $\mathcal{R}_{\text{spectral}}$ and $\mathcal{L}_{\text{Lyap}}$ shape the transition operator $\mathbf{K}$ without distorting encoder representations.

Proposition 1 (Operator Non-Triviality Under Finite-ϵ and Singular Barriers)**.** *Consider* $\mathcal{L}_\epsilon(\mathbf{K}) = \frac{1}{BD}\sum_{b=1}^B \|\mathbf{K}z_{ctx}^{(b)} - z_{tgt}^{(b)}\|_2^2 + \lambda_{ent}\mathcal{R}_{spectral}(\mathbf{K})$ *with* $\lambda_{ent}, \beta_{norm} > 0$ *and empirical context–target cross-covariance* $\mathbf{M} = \frac{1}{BD}\sum_{b=1}^B z_{tgt}^{(b)} z_{ctx}^{(b)\top}$*: (i) Under the finite-*ϵ *smooth barrier (*$\epsilon = 10^{-6}$*), the regularizer gradient vanishes at the origin,* $\nabla_{\mathbf{K}}\mathcal{R}_{spectral}(\mathbf{0}) = \mathbf{0}$*, because* $r(\mathbf{K})$ *is quadratic in* $\mathbf{K}$*; however, whenever context and target embeddings exhibit non-zero cross-covariance (*$\mathbf{M} \neq \mathbf{0}$*),* $\nabla_{\mathbf{K}}\mathcal{L}_\epsilon(\mathbf{0}) = -2\mathbf{M} \neq \mathbf{0}$*, which excludes* $\mathbf{K} = \mathbf{0}$ *as a stationary point or minimizer. (ii) In the singular-barrier limit (*$\epsilon \to 0^+$*),* $\lim_{\|\mathbf{K}\|_F \to 0} \Phi_{energy}(\mathbf{K}) = +\infty$ *unconditionally.*

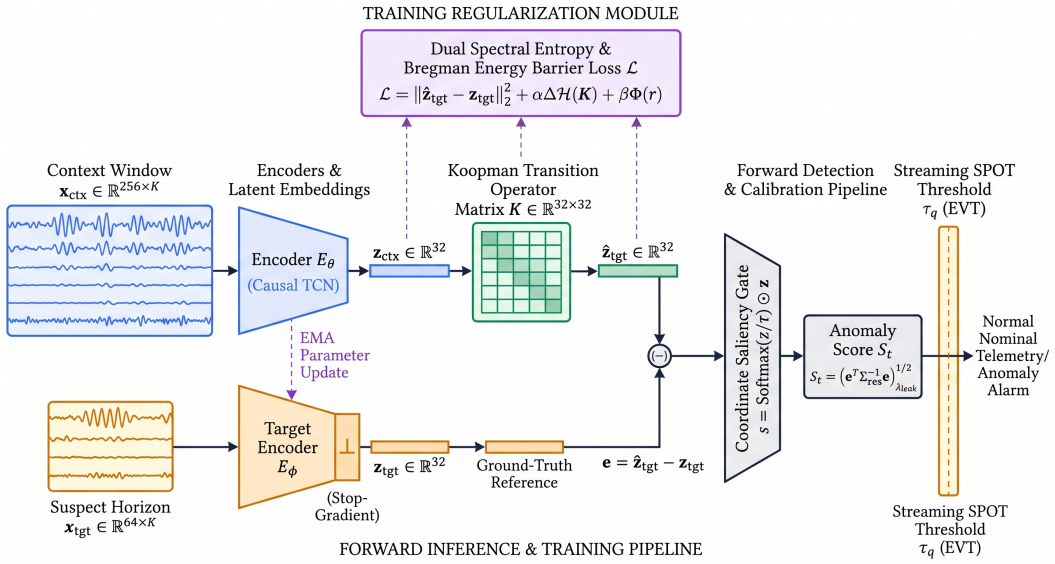


Fig. 1. Operator-Entropy JEPA architecture. Online (E_θ) and momentum target (E_ϕ) encoders map context ($\mathbf{x}_{\text{ctx}} \in \mathbb{R}^{256\times K}$) and suspect ($\mathbf{x}_{\text{tgt}} \in \mathbb{R}^{64\times K}$) windows into latent space, where a Koopman operator ($\mathbf{K} \in \mathbb{R}^{32\times32}$) forecasts $\hat{\mathbf{z}}_{\text{tgt}} = \mathbf{K}\mathbf{z}_{\text{ctx}}$ under spectral-entropy and Bregman-barrier regularization for coordinate-gated SPOT scoring.

Proof. For $\epsilon > 0$, $\nabla_{\mathbf{K}}\Phi_{\text{energy}}(\mathbf{K}) = \frac{2}{D}\Phi'_{\text{energy}}(r)\,\mathbf{K} = \mathbf{0}$ at $\mathbf{K} = \mathbf{0}$, and $\mathbf{K} = \mathbf{0}$ yields uniform smoothed probabilities $p_i = 1/D$, minimizing $\Delta\mathcal{H} \geq 0$ so $\nabla_{\mathbf{K}}\Delta\mathcal{H}(\mathbf{0}) = \mathbf{0}$. Differentiating the prediction term gives $\nabla_{\mathbf{K}}\mathcal{L}_\epsilon(\mathbf{0}) = -2\mathbf{M} \neq \mathbf{0}$ whenever $\mathbf{M} \neq \mathbf{0}$. As $\epsilon \to 0^+$, $-\log r(\mathbf{K}) \to +\infty$ as $\|\mathbf{K}\|_F \to 0$. □

Remark (Scope of Regularization): Proposition 1 guarantees non-degeneracy of the transition matrix $\mathbf{K}$, not diversity of the encoder representations $z_{\text{ctx}}, z_{\text{tgt}}$. Unlike representation regularizers such as VICReg [20] or Barlow Twins [18] that penalize encoder covariance collapse directly, $\mathcal{R}_{\text{spectral}}(\mathbf{K})$ constrains only the transition spectrum; encoder collapse is prevented empirically by the momentum target encoder and stop-gradient decoupling (Table I(c)).

2) Comparative Physical Priors (Reynolds-Stress and Potential-Flow Heads): To test whether alternative continuum-mechanics priors improve anomaly detection, we evaluate two additional latent predictor heads under identical encoder backbones:

- **Reynolds-Stress JEPA (Second-Moment Closure Head):** Inspired by Reynolds decomposition, this variant augments the mean latent prediction $\hat{z}_{\text{tgt}}$ with a Cholesky factor head $\mathbf{L} \in \mathbb{R}^{d_s\times d_s}$ ($d_s = 16$) that predicts the projected outer-product second moment $\boldsymbol{\Sigma}_{\text{obs}} = (\mathbf{P}z_{\text{tgt}})(\mathbf{P}z_{\text{tgt}})^\top + 10^{-3}\mathbf{I}$ via $\boldsymbol{\Sigma}_{\text{pred}} = \mathbf{L}\mathbf{L}^\top + 10^{-3}\mathbf{I}$. Training minimizes the sum of the latent prediction MSE and $0.1\,\text{MSE}(\boldsymbol{\Sigma}_{\text{pred}}, \boldsymbol{\Sigma}_{\text{obs}})$, and inference adds $0.5\,g(\text{vec}(\boldsymbol{\Sigma}_{\text{obs}} - \boldsymbol{\Sigma}_{\text{pred}}))$ to the primary residual score.
- **Potential-Flow JEPA (Curl-Free Flow Matching Head):** Motivated by conservative vector fields, this variant parameterizes a curl-free velocity field $\hat{v} = -\nabla_{z_t}\Phi_\psi$ via a residual MLP conditioned on flow time $t \in [0, 1]$, context z_{ctx}, and a soft router over $K_r = 4$ orthonormal subspaces. Training uses conditional flow matching along linear paths from Gaussian noise to z_{tgt}, and inference evaluates the velocity residual at the trajectory midpoint. Because conservative gradient fields penalize rotational trajectories, this bias is hypothesized to degrade detection on periodic telemetry (Section VII-B).

C. Anomaly Scoring and Operational Detection Latency

At test time, sliding windows are processed at stride 1. Let $\mathbf{e}(t) = z_{\text{tgt}}(t) - \hat{z}_{\text{tgt}}(t) \in \mathbb{R}^D$ denote the latent prediction residual, and let $g(\mathbf{e})$ denote the coordinate saliency gate defined in (5). Unconstrained `TS-JEPA` scores windows directly via $\mathcal{S}(t) = g(\mathbf{e}(t))$. Because the final LayerNorm layer projects latent vectors onto the sphere of radius $\sqrt{D}$ and strips radial amplitude, Operator-Entropy JEPA records the pre-LayerNorm radial energy scaled by window RMS for both encoders ($e_{\text{ctx}}, e_{\text{tgt}}$), forms the predicted radial energy $e_{\text{pred}} = e_{\text{ctx}}\|\hat{z}_{\text{tgt}}\|_2/(\|z_{\text{ctx}}\|_2 + 10^{-6})$, and combines directional and radial discrepancies:

$$\mathcal{S}_{\text{OpEnt}}(t) = g\big(\mathbf{e}(t)\big) + 0.2\,\frac{|e_{\text{tgt}}(t) - e_{\text{pred}}(t)|}{e_{\text{tgt}}(t) + e_{\text{pred}}(t) + 10^{-6}}. \quad (4)$$

Three-Part Operational Causality Framework: To avoid confounding data access, window buffering, and alarm timing, we formalize online evaluation under three criteria (illustrated in Fig. 2):

(a) **Data causality ($\mathcal{F}_\tau$-measurability):** At decision timestamp $\tau = t + S - 1$, the detector observes only historical data $\mathbf{X}[t - C : \tau, :]$; neither encoder accesses observations beyond τ.

(b) **Score-emission latency ($S - 1 = 63$ steps) vs. fault-detection delay (Δt):** Because z_{tgt} encodes the S-step target block $\mathbf{X}[t : t + S, :]$, window score s_t is emitted at the trailing edge $\tau = t + S - 1$ (Fig. 2, Row 3). This creates a structural score-emission latency of $S - 1 = 63$ steps (0.98 s on 64 Hz Daphnet, 10.5 min on 0.1 Hz SMAP/MSL/GHL, 63 min on GECCO, and 63 h on hourly Metro). This structural lag is distinct from empirical fault-detection delay $\Delta t = t_{\text{alarm}} - t^*$, which measures the elapsed steps from ground-truth anomaly

onset t^* to the first timestamp $t_{\text{alarm}} \geq t^*$ whose attributed score exceeds threshold τ_{SPOT}. Shortening the horizon to $S \in \{1, 4\}$ eliminates score-emission latency when rapid tripwire response is required (Section VII-D).

(c) **Strictly causal trailing vs. retrospective buffered attribution:** Under *strictly causal trailing attribution*, window score s_t is assigned only to the decision timestamp $\tau = t + S - 1$ for zero-lookahead streaming. Under *retrospective buffered smear attribution*, s_t is averaged across all steps in $[t + 1, t + S]$, which introduces fixed-lag smoothing and can smear post-onset scores up to S steps before t^*. Both regimes are reported separately throughout (Table II(a),(b)).

D. Shared Architectural Components and Thresholding

1) Coordinate Saliency Gating: In high-dimensional streams ($K \geq 25$), faults affecting a single subsystem can be diluted by nominal variance across the remaining coordinates. To amplify sparse latent deviations, we apply a softmax coordinate saliency gate $g(\mathbf{e}) \equiv \mathcal{D}_{\text{sal}}(\mathbf{e})$ to the residual vector $\mathbf{e} = \hat{z}_{\text{tgt}} - z_{\text{tgt}} \in \mathbb{R}^D$:

$$\mathcal{D}_{\text{sal}}(\mathbf{e}) = (1-\alpha)\|\mathbf{e}\|_2 + \alpha\sqrt{D\sum_{d=1}^{D} w_d e_d^2}, \quad w_d = \frac{\exp(|e_d|/\tau_g)}{\sum_{j=1}^{D}\exp(|e_j|/\tau_g)} \tag{5}$$

with fixed blend $\alpha = 0.5$ and temperature $\tau_g = 0.5$ across all latent models. Input channels are encoded with uniform weights in the deployed pipeline; an auxiliary context-variance input-channel gate is evaluated in Section VI-C.

2) Multiscale Frequency Front-End: Before the dilated TCN backbone, each input channel is split across $J = 4$ smooth dyadic frequency shells on a 256-point rFFT grid normalized into an exact amplitude partition of unity ($\sum_{j=1}^{J} \Psi_j[\omega_m] \equiv 1$ to within 1.2×10^{-7}). Each shell is processed by a causal 1D convolution with kernel sizes $\{15, 9, 5, 3\}$ from coarse to fine scales and recombined via learned softmax routing. Because smooth Fourier shells overlap at band edges, this module acts as a multiscale frequency front-end rather than a strictly orthogonal wavelet bank (Table I(b)).

3) Extreme-Value Threshold Calibration (SPOT): To convert continuous scores $\mathcal{S}(t)$ into binary alarms $\hat{y}_t = \mathbb{I}(\mathcal{S}(t) > \tau_{\text{SPOT}})$ using nominal calibration data alone, we apply Streaming Peaks-Over-Threshold (SPOT) [7]. Score excesses above the 98th calibration percentile $u = p_{98}$ are fit to a Generalized Pareto Distribution (GPD) with shape ξ and scale σ via Grimshaw maximum likelihood, yielding the threshold:

$$\tau_{\text{SPOT}} = u + \frac{\sigma}{\xi}\left[\left(\frac{qN_{\text{cal}}}{N_u}\right)^{-\xi} - 1\right], \tag{6}$$

where $q = 10^{-3}$ is the nominal design risk, N_{cal} is the number of calibration windows, and N_u is the exceedance count above u. All primary benchmarks use standard unadjusted SPOT (τ_{SPOT}). In Section VII-F, we also audit two heuristic kurtosis adjustments: total-threshold scaling $\tau_{\text{eff}} = \tau_{\text{SPOT}}[1 + \frac{1}{6}\tanh(\gamma_2)]$ and excess-only scaling $\tau_{\text{exc}} = u + (\tau_{\text{SPOT}} - u)[1 + \frac{1}{6}\tanh(\gamma_2)] \geq u$.

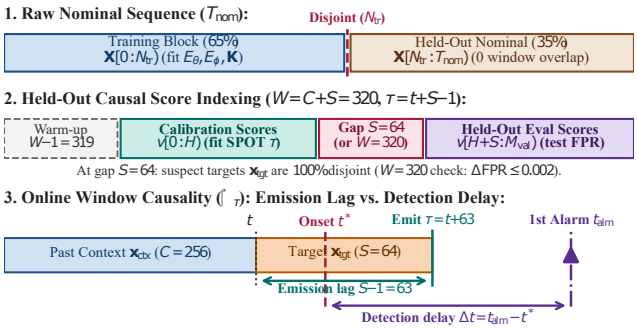


Fig. 2. Chronological split, boundary window indexing, and online causality protocol (C=256, S=64, W=320). Row 1 separates raw training (65%) and held-out nominal (35%) observations with zero window overlap across N_{tr}. Row 2 indexes causal held-out scores $v[0 : M_{\text{val}})$ into context warm-up ($W-1$=319), calibration $v[0 : H)$, isolation gap S=64 (or full-window W=320), and held-out evaluation $v[H+S : M_{\text{val}})$. Row 3 distinguishes structural score-emission lag ($S-1$=63) from empirical fault-detection delay ($\Delta t = t_{\text{alm}} - t^*$).

V. Experimental Protocol and Evaluation Metrics

A. Experimental Setup and Hyperparameters

All architectures are trained and evaluated under an identical streaming time-series pipeline using a past context window of $C = 256$ steps and a suspect target horizon of $S = 64$ steps (total window length $W = C + S = 320$), evaluated at stride 1. Context and target representations ($D = 32$) are extracted by the multiscale frequency front-end followed by a 6-block causal dilated TCN (48 filters, kernel size 5, dilations $\{1, 2, 4, 8, 16, 32\}$). Multi-statistic temporal pooling (terminal, mean, max, and channel statistics) projects features through a linear bottleneck before LayerNorm and dropout ($p = 0.20$). Models are trained for 20 epochs on nominal sequences using AdamW (initial learning rate 10^{-3} decayed to 10^{-5} via cosine annealing, weight decay 10^{-4}, batch size 32).

a) Held-Out Calibration Protocol and Boundary Isolation: To evaluate out-of-sample threshold calibration without circularity, each stream's raw nominal sequence $\mathbf{X}_{\text{nom}} \in \mathbb{R}^{T_{\text{nom}} \times K}$ is partitioned chronologically (Fig. 2, Row 1) into a *training block* $\mathbf{X}_{\text{tr}} = \mathbf{X}_{\text{nom}}[0 : N_{\text{tr}})$ ($N_{\text{tr}} = \lfloor 0.65\, T_{\text{nom}} \rfloor$) and a disjoint *held-out nominal block* $\mathbf{X}_{\text{val}} = \mathbf{X}_{\text{nom}}[N_{\text{tr}} : T_{\text{nom}})$. Standardization parameters and neural weights are fit on $\mathbf{X}_{\text{tr}}$ windows alone, guaranteeing zero raw-observation or window overlap across N_{tr}.

Within $\mathbf{X}_{\text{val}}$ ($N_{\text{val}} = T_{\text{nom}} - N_{\text{tr}}$), sliding windows of length $W = 320$ are constructed strictly inside $\mathbf{X}_{\text{val}}$ (Fig. 2, Row 2). Under causal trailing attribution, the first $W-1 = 319$ observations serve as context warm-up, yielding $M_{\text{val}} = N_{\text{val}} - W + 1$ valid score timestamps $v[0 : M_{\text{val}})$. Splitting v at midpoint $H = \lfloor M_{\text{val}}/2 \rfloor$ defines the *nominal calibration block* $v_{\text{cal}} = v[0 : H)$ (used to fit the SPOT threshold at $u = p_{98}, q = 10^{-3}$) and the *held-out evaluation block* $v_{\text{hold}} = v[H + S : M_{\text{val}})$, excluding $S = 64$ intermediate windows $v[H : H + S)$. Because suspect target windows have length $S = 64$, the last calibration target and first evaluation target are 100% disjoint; the first $C = 256$ evaluation steps share passive historical context with calibration observations, mirroring online streaming where past nominal history forms context for current predictions. A stricter full-window isolation check ($G = W = 320$, excluding

$v[H : H + 320)$ so even passive context is disjoint across the seven streams with $M_{\text{val}}/2 > 420$) leaves stream-macro causal FPR virtually unchanged ($2.07\% \rightarrow 2.18\%$ for OpEntropy and $3.02\% \rightarrow 3.24\%$ for TS-JEPA versus $12.15\% \rightarrow 13.97\%$ for TimesNet and $9.05\% \rightarrow 9.73\%$ for TranAD). All primary tables use fixed standard SPOT thresholds during testing.

b) Multi-Seed Replication and Equivalence Testing: Principal models on the 19 non-GHL streams are evaluated across five random seeds ($\{42, 123, 456, 7, 2024\}$); the 14 GHL streams ($> 10^6$ observations each) and ablation controls use three seeds. Seeds are averaged within each stream before computing stream-resampled paired bootstrap confidence intervals (2,000 iterations) and Benjamini–Hochberg FDR-adjusted Wilcoxon signed-rank tests ($q_{\text{FDR}} = 0.05$).

To test whether physics-inspired regularizers alter detection relative to unregularized controls, we pair two-sided tests with Two One-Sided Tests (TOST) of equivalence: $H_{01} : \mu_1 - \mu_2 \leq -\Delta$ (p_{lower}) and $H_{02} : \mu_1 - \mu_2 \geq +\Delta$ (p_{upper}), with overall $p_{\text{TOST}} = \max(p_{\text{lower}}, p_{\text{upper}})$ and 90% confidence intervals. We prespecify $\Delta = \pm 0.03$ Point-F1 for fixed-backbone regularizer controls ($\approx 13\%$ of baseline F1, $< 0.09\times$ between-stream SD) and $\Delta = \pm 0.05$ for cross-architecture comparisons ($\approx 25\%$ of baseline F1, corresponding at prevalence $\pi \approx 0.15$ to an event-recall shift of ≤ 0.04 or 15–20 false alarms per 1,000 nominal steps), and additionally evaluate narrow-margin TOST sensitivity down to $\Delta \in \{\pm 0.010, \pm 0.015, \pm 0.020\}$ (Table I(a)). Formal equivalence requires both $p_{\text{TOST}} < 0.05$ and the 90% CI strictly inside $[-\Delta, +\Delta]$.

B. Baseline Architectures, Fairness Checks, and Cohort Structure

We separate primary multi-seed conclusions (Cohorts 1–2 across 12 neural, linear, and non-parametric detectors) from secondary single-seed cross-checks. Three fairness audits verify that reconstructive baseline results are not artifacts of window length or training budget:

1) **Training-budget sensitivity** (7 streams $\times 3$ seeds): Doubling training from 10 to 20 epochs slightly improves JEPA Point-F1 ($0.038 \rightarrow 0.048$ for TS-JEPA; $0.029 \rightarrow 0.065$ for OpEntropy) while reconstructor Point-F1 plateaus ($0.154 \rightarrow 0.151$ for TimesNet; $0.161 \rightarrow 0.158$ for TranAD) and held-out FPR remains unchanged (8.7%–9.2% vs. 33.3%).
2) **Native published window lengths** (5 streams): Evaluating TranAD ($W = 10$) and TimesNet ($W = 96$) at their published window sizes raises Point-F1 (0.164 and 0.218), but observation residuals remain severely leptokurtic ($\gamma_2 =$ 448.3 and 98.1 vs. 3.1 for TS-JEPA) with high nominal FPR (28.4% and 44.1%).
3) **Secondary 130-stream single-seed cross-check:** Reconstructors replicate their published point-adjusted F1 on SMD and MSL, and Table II(d) provides a secondary 130-stream, 11-dataset single-seed comparison against NCAD, DCdetector, and Anomaly Transformer.

C. Cohort Summary and Reproducibility Specifications

Experiments are organized into five non-overlapping cohorts:

- **Cohort 1 (33-stream primary multi-seed benchmark):** 19 non-GHL streams (5 seeds) and 14 GHL streams (3 seeds), reported in Tables II(a)–(c) and IV(a).
- **Cohort 2 (19-stream mechanism and operating-curve suite):** 12 controlled backbone arms across 5 seeds, reported in Table I(a),(e), Table III(b), and Fig. 3.
- **Cohort 3 (6-stream operator sweep and quiet-fault suite):** Factorial regularizer sweep (3 seeds) and six injected single-channel fault types, reported in Table I(c),(d).
- **Cohort 4 (5-stream horizon and frequency-shell panels):** Target-horizon latency sweep ($S \in \{1, 4, 16, 32, 64\}$, 3 seeds) and dyadic front-end ablation, reported in Table III(a) and Table I(b).
- **Cohort 5 (130-stream single-seed baseline screen):** Secondary cross-check across 11 datasets against NCAD, DCdetector, and Anomaly Transformer (Table II(d)).

At $K = 25$ channels, model parameter counts are comparable across families: 285,701 (OpEntropy JEPA), 289,061 (TS-JEPA), 296,037 (Reynolds-Stress), 338,209 (Potential-Flow), 648,553 (TimesNet), and 205,746 (TranAD). Non-finite telemetry entries are zero-imputed with channel standard deviations floored at 10^{-2}; SPOT fits GPD parameters (ξ, σ) above $u = p_{98}$ via Grimshaw MLE with L-BFGS-B fallback ($\xi \in [-0.9, 10], \sigma \geq 10^{-6}$). All 195 primary multi-seed runs converged on PyTorch 2.4 / CUDA 12.4 (NVIDIA RTX 5090, 0.60–0.84 ms per window, 184 MB VRAM); stream manifests, fitted thresholds, and source code are available at https://github.com/quaid281/NCAD.

D. Multi-Tier Statistical Evaluation Metrics

To guard against point-adjustment inflation, channel-averaging bias, and threshold gaming [28], [36], we evaluate models across three complementary tiers: (i) *primary calibration and accuracy endpoints*, comprising held-out nominal FPR, unadjusted Point-F1, and threshold-free PR-AUC; (ii) *operating characteristics*, evaluating event recall across deployable calibration quantiles (q^*, Table III(b)) and across post-hoc matched realized test-FPR budgets ($f^* \in [0.01, 0.50]$, Fig. 3(b)); and (iii) *secondary operational endpoints*, reporting unweighted Dataset-Macro Point-F1 (preventing the 14-stream GHL block from dominating), event recall (fraction of contiguous anomaly segments with ≥ 1 alarm), mean detection delay Δt over detected events, and the constant-alarm baseline $\text{F1}_{\text{all-pos}} = 2\pi/(1 + \pi)$ at anomaly prevalence π.

VI. Ablation Studies and Analysis

A. Ablation 1: Physical Regularizer Controls on Fixed Architectures

To test whether physics-inspired regularizers drive anomaly detection accuracy, Table I(a) disables each physical penalty on its fixed architecture across all 19 non-GHL streams:

1) **Operator-Entropy vs. λ_{ent}=0 (fixed Koopman backbone):** Removing the spectral regularizer leaves buffered Point-F1 (0.232 vs. 0.235, paired difference -0.004, two-sided $p = 0.89$) and PR-AUC (0.365 vs. 0.365) virtually identical. Two One-Sided Tests confirm formal equivalence

TABLE I
ABLATION AND MECHANISM-DISCRIMINATION STUDIES: (A) FIXED-BACKBONE REGULARIZER CONTROLS AND TOST EQUIVALENCE (n=19, 5 SEEDS), (B) DYADIC FRONT-END (n=5), (C) OPERATOR REGULARIZER SWEEP (n=6, 3 SEEDS), (D) INJECTED QUIET-FAULT POINT RECALL (n=6), AND (E) 12-ARM BACKBONE COMPARISON (n=19, 5 SEEDS).

(a) Regularizer Controls on Fixed Architectures & TOST Equivalence

Model / Control	Point-F1	PR-AUC	FPR	ER	Paired ΔF1 (90% CI)	p_{TOST}
TS-JEPA (Unconstrained)	0.206 ± 0.025	0.331	0.093	0.477	Baseline (cross-arch)	–
Operator-Entropy JEPA	0.233 ± 0.028	0.351	0.101	0.550	-0.004 $[-0.011, +0.003]$	0.0007 ($\pm$.03)
$\lambda_{\text{ent}} = 0$ control	0.235 ± 0.038	0.365	0.099	0.580	(also $\pm$.015: p=0.0024)	0.0088 ($\pm$.01)
Reynolds-Stress JEPA	0.190 ± 0.022	0.313	0.074	0.457	$+0.013$ $[-0.001, +0.030]$	0.0019 ($\pm$.05)
$\alpha_{\text{stress}} = 0$ control	0.182 ± 0.036	0.299	0.067	0.472	(also $\pm$.035: p=0.0031)	–
Potential-Flow JEPA	0.138 ± 0.044	0.235	0.087	0.449	-0.054 $[-0.131, +0.012]$	0.17 (fails)
no-regime control ($K_r = 1$)	0.189 ± 0.024	0.279	0.084	0.431	–	–
TimesNet (Baseline)	0.197 ± 0.018	0.329	0.300	0.583	–	–
TranAD (Baseline)	0.130 ± 0.004	0.287	0.307	0.490	–	–

(b) Dyadic-Shell Front-End On/Off

Stream (F1 / PR-AUC)	Standard Conv	Dyadic Shells	ΔF1 / ΔPR
SMAP `P-3`	**0.301** / 0.280	0.115 / **0.296**	-0.186 / $+0.017$
MSL `M-1`	0.014 / **0.491**	**0.016** / 0.449	$+0.002$ / -0.043
SMD `machine-1-2`	0.000 / 0.028	**0.068** / **0.221**	$+0.068$ / $+0.193$
Daphnet `S01R01E1`	**0.179** / **0.103**	0.130 / 0.066	-0.048 / -0.037
GECCO `water_quality`	**0.097** / 0.161	0.062 / **0.282**	-0.035 / $+0.121$
Macro Mean	**0.118** / 0.213	0.078 / **0.263**	-0.040 / $+0.050$

(c) Factorial and Strength Sweep of the Operator Regularizer

			Operator / representation / stability				Buffered			Causal		
Arm	λ_{ent}	β_{norm}	RankMe($\mathbf{K}$)	$\kappa(\mathbf{K})$	RankMe($z_{\text{ctx}}/z_{\text{tgt}}$)	$\text{CV}_{\mathcal{L}}$	FPR	F1	ER	FPR	F1	ER
Neither	0	0	31.53	1.437	5.16/11.57	0.565	0.071	0.044	0.250	0.017	0.009	0.369
Entropy only	0.5	0	31.84	1.256	5.25/11.57	0.550	0.071	0.040	0.266	0.017	0.011	0.389
Barrier only	0	1	31.54	1.431	5.37/11.58	0.563	0.074	0.047	0.266	0.016	0.011	0.389
Both	0.05	1	31.61	1.395	5.16/11.57	0.562	0.074	0.039	0.250	0.017	0.008	0.373
Both	0.10	1	31.67	1.363	5.19/11.58	0.559	0.073	0.039	0.250	0.017	0.008	0.421
Both (deployed)	0.50	1	31.85	1.255	5.54/11.56	0.549	0.072	0.038	0.266	0.017	0.010	0.444
Both	1.00	1	31.91	1.204	5.25/11.55	0.546	0.070	0.037	0.250	0.017	0.007	0.365
Both	2.00	1	31.95	1.154	5.34/11.54	0.545	0.072	0.032	0.250	0.017	0.004	0.397

(d) Quiet-Fault Point Recall

Injected fault	Unif.	Var.	MAD	`lowvar`	`dev_maxz`
Flatline	0.021	0.043	0.058	0.182	0.087
Flatline (mid-var.)	0.024	0.048	0.068	0.317	0.087
Stuck sensor	0.021	0.043	0.058	0.182	0.087
Dropout	0.020	0.043	0.058	0.182	0.087
Low-amp. step	0.030	0.049	0.062	0.182	0.087
$+5\sigma$ spike	0.023	0.042	0.059	0.182	0.108
Mean nom. FPR	0.019	0.069	0.055	0.149	0.141

(e) Controlled Mechanism-Discrimination Study (19 Non-GHL Streams)

	Tail/Drift		Held FPR		Point-F1		PR-AUC		Causal Event Recall			
Arm	γ_2	D_{KS}	Caus.	Buf.	Caus.	Buf.	Caus.	Buf.	SPOT	q^*=0.05	f^*=0.01	f^*=0.05
`TS-JEPA`	1.33	0.214	0.024	0.093	0.161	0.206	0.278	0.331	0.487	0.730	0.395	0.584
`OpEntropy`	0.94	0.209	0.028	0.101	0.153	0.233	0.280	**0.351**	0.517	0.752	0.371	0.573
`JEPA-CtxRecon`	2.59	0.203	0.017	0.108	0.112	0.188	0.235	0.256	0.542	0.795	**0.431**	0.573
`JEPA-NoEMA`	1.16	**0.167**	**0.009**	0.098	0.146	**0.235**	0.273	0.328	0.503	**0.816**	0.398	0.600
`TS-JEPA-NoGate`	0.91	0.214	0.018	0.077	0.156	0.209	0.272	0.324	0.538	0.728	0.418	0.599
`PCA-LinPred-LN`	0.95	0.233	0.021	0.186	0.017	0.104	0.210	0.244	0.452	0.665	0.377	**0.624**
`TCN-ObsPred-LN`	0.91	0.376	0.117	0.193	0.084	0.126	0.235	0.257	0.421	0.665	0.329	0.541
`IsoForest`	**0.70**	0.373	0.039	**0.056**	0.041	0.041	0.209	0.222	0.149	0.631	0.317	0.443
`TCN-ObsPred`	6.10	0.463	0.247	0.313	0.140	0.145	0.272	0.310	0.546	0.629	0.171	0.383
`TCN-ObsRecon`	6.24	0.445	0.245	0.304	0.134	0.135	0.261	0.301	0.528	0.619	0.163	0.379
`TimesNet`	5.54	0.438	0.243	0.300	**0.209**	0.197	**0.293**	0.329	**0.588**	0.618	0.217	0.407
`TranAD`	5.64	0.459	0.222	0.307	0.140	0.130	0.248	0.287	0.533	0.602	0.149	0.343

within the prespecified $\Delta = \pm 0.03$ margin ($p_{\text{TOST}} = 0.0007$) and across narrow-margin sensitivity checks down to $\Delta = \pm 0.020$ ($p_{\text{TOST}} = 0.0015$), $\Delta = \pm 0.015$ ($p_{\text{TOST}} = 0.0024$, with 90% CI $[-0.0112, +0.0029]$ strictly inside $[-\Delta, +\Delta]$), and $\Delta = \pm 0.010$ ($p_{\text{TOST}} = 0.0088$), as well as within ± 0.010 PR-AUC and ± 0.025 event recall.

2) **Reynolds-Stress vs. α_{stress}=0 (fixed backbone):** Disabling the second-moment loss yields a $+0.013$ Point-F1 shift (two-sided $p = 0.59$), confirming TOST equivalence within $\Delta = \pm 0.05$ ($p_{\text{TOST}} = 0.0019$) and down to $\Delta = \pm 0.035$ ($p_{\text{TOST}} = 0.0031$).
3) **Potential-Flow vs. no-regime (K_r=1):** Disabling multi-regime routing shifts Point-F1 by -0.054 ($p_{\text{TOST}} = 0.17$), failing equivalence at $\Delta = \pm 0.05$.

These fixed-backbone regularizer controls must be distinguished from cross-architecture comparisons against unconstrained TS-JEPA (which uses an MLP predictor without radial energy scoring or Lyapunov covariance alignment). On the 19-stream subset, the $+0.028$ Point-F1 difference between OpEntropy JEPA and TS-JEPA (90% CI $[-0.018, +0.074]$, $p_{\text{TOST}} = 0.11$) is statistically inconclusive at $\Delta = \pm 0.05$, whereas across all 33 primary streams OpEntropy JEPA ($+0.016$, 90% CI $[-0.009, +0.045]$, $p_{\text{TOST}} = 0.0019$) and Reynolds-Stress (-0.008, $p_{\text{TOST}} = 0.0017$) satisfy TOST(± 0.05) (with OpEntropy vs. TS-JEPA also satisfying TOST(± 0.030) at $p_{\text{TOST}} = 0.0077$ and ± 0.015 causal FPR at $p_{\text{TOST}} = 0.0001$).

B. Ablation 2: Factorial Isolation of the Entropy Term and Energy Barrier

Table I(c) crosses the spectral-entropy term and Bregman energy barrier across $\lambda_{\text{ent}} \in [0, 2.0]$ on six streams (three seeds). Three findings emerge. First, because the Koopman matrix $\mathbf{K}$ is initialized near the identity, the unregularized operator is already near full rank ($\text{RankMe}(\mathbf{K}) = 31.53$ of 32, condition number $\kappa = 1.437$); increasing λ_{ent} tightens conditioning monotonically ($\kappa = 1.255$ at the deployed $\lambda_{\text{ent}} = 0.50$, and $\kappa = 1.154$ at $\lambda_{\text{ent}} = 2.00$) while lowering training loss variation ($\text{CV}_{\mathcal{L}}$: $0.565 \rightarrow 0.545$). Second, because $\mathcal{L}_{\text{Lyap}}$ applies a stop-gradient to z_{ctx}, encoder effective ranks remain invariant across all regularizer strengths ($\text{RankMe}(z_{\text{ctx}}) \in [5.16, 5.54]$, $\text{RankMe}(z_{\text{tgt}}) \in [11.54, 11.58]$). Third, detection and false-alarm rates remain stable across arms, with the deployed setting ($\lambda_{\text{ent}} = 0.50, \beta_{\text{norm}} = 1$) attaining the highest causal event recall (0.444 vs. 0.369 unregularized). On an eight-stream decoupling panel, pairing detached Lyapunov alignment ($\beta_{\text{Lyap}} = 0.02$) with the smooth Bregman barrier cuts causal detection delay from 127.0 steps (TS-JEPA) and 80.2 steps ($\beta_{\text{Lyap}} = 0$) to 58.2 steps while raising causal Point-F1 from 0.044 to 0.094.

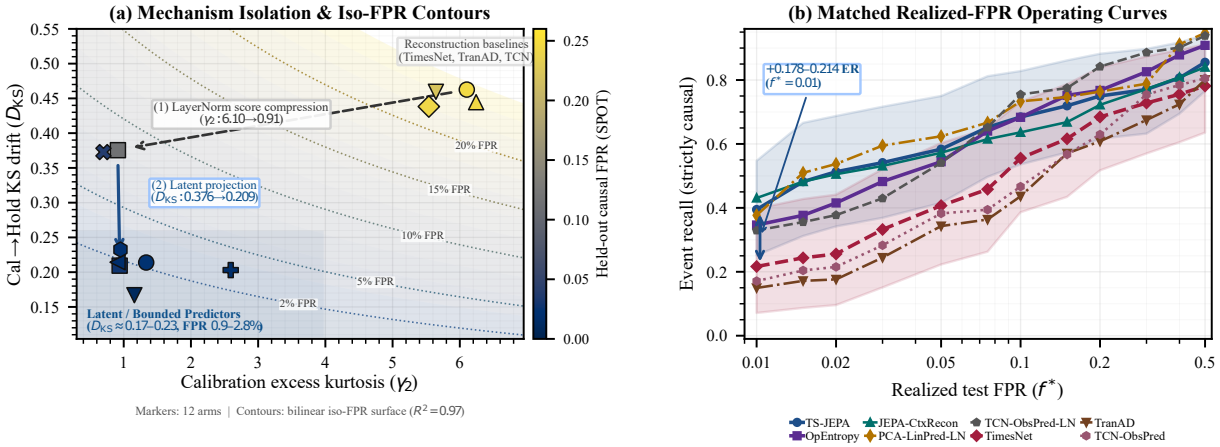


Fig. 3. Residual conditioning and operating curves across 19 non-GHL streams (five seeds, strictly causal trailing inference). **(a)** Calibration excess kurtosis (γ_2) vs. calibration-to-holdout KS drift (D_{KS}), colored by held-out causal FPR at SPOT q=10^{-3} with bilinear iso-FPR topographic contours (R^2=0.97), illustrating LayerNorm score compression and learned latent projection. **(b)** Causal event recall across matched realized test-FPR budgets $f^* \in [0.01, 0.50]$ with 95% stream-bootstrap confidence bands.

C. Ablation 3: Auxiliary Input-Channel Gating and Quiet-Fault Monitors

Comparing the auxiliary context-variance input-channel gate against uniform channel weighting on four streams confirms that uniform weighting achieves higher macro Point-F1 (0.118 vs. 0.077), so the deployed pipeline retains uniform input encoding. In Table I(d), injecting six single-channel fault types into six multivariate streams shows that quiet faults (flatlines, stuck sensors, dropouts) yield low point recall (0.02–0.07) across all latent gating variants because LayerNorm embeddings absorb constant single-channel offsets as nominal background. Auxiliary observation-space monitors, namely a rolling variance floor (`lowvar`) and maximum channel z-score (`dev_maxz`), recover quiet-fault recall (0.18–0.32) at the cost of inflating nominal FPR eight-fold (14.1%–14.9% vs. 1.9%).

D. Ablation 4: Multiscale Dyadic-Shell Front-End

In Table I(b), replacing the standard convolutional stem with the 4-shell Fourier front-end increases macro PR-AUC from 0.213 to 0.263 (+0.050, led by SMD and GECCO) and lowers calibration residual kurtosis (1.409 → 1.069), while macro Point-F1 shifts from 0.118 to 0.078 due to conservative threshold tightening on SMAP `P-3`. The module thus acts as a ranking-conditioning stem rather than a uniform F1 gain.

E. Mechanism-Discrimination Study: Latent Space vs. Observation Space

Comparing the 12 controlled backbone arms across all 19 non-GHL streams in Table I(e) and Fig. 3(a) separates geometric score compression from representation drift reduction:

1) **Unnormalized observation forecasting (`TCN-ObsPred`):** Raw observation residuals exhibit heavy tail kurtosis (γ_2 = 6.10), severe calibration-to-holdout KS drift (D_{KS} = 0.463), and high causal false alarms (24.7%).
2) **LayerNorm observation bottleneck (`TCN-ObsPred-LN`):** Inserting a D = 32 LayerNorm bottleneck before decoding mechanically bounds excursions, compressing kurtosis to γ_2 = 0.91 and halving causal FPR to 11.7%, but leaves calibration-to-holdout drift elevated (D_{KS} = 0.376).
3) **Learned latent projection (`TS-JEPA`, `OpEntropy`):** Forecasting in learned latent coordinates halves calibration-to-holdout KS drift (D_{KS} : 0.376 → 0.214 and 0.209, paired Wilcoxon $p \leq 0.001$) and reduces causal false alarms an additional five- to eight-fold (1.7%–2.8%).
4) **Context autoencoding (`JEPA-CtxRecon`) and linear controls:** Autoencoding context embeddings without temporal foresight achieves comparable drift reduction (D_{KS} = 0.203, causal FPR 1.7%, event recall 0.542), while linear projection (`PCA-LinPred-LN`) reaches 2.1% causal FPR at lower F1 (0.017) and `IsoForest` collapses in recall (0.149). LayerNorm geometry thus bounds tail kurtosis, while learned latent projection is required to suppress temporal distribution drift.

VII. Results and Discussion

A. Benchmark Performance on Continuous Telemetry

Table II summarizes performance across all 33 continuous telemetry streams under the chronological held-out SPOT calibration protocol ($u = p_{98}, q = 10^{-3}$), reporting retrospective buffered attribution in Panel (a), strictly causal streaming in Panel (b), and per-dataset breakdowns in Panel (c). The central empirical finding is substantial relative false-alarm suppression under automated thresholding, whereas overall Point-F1 and PR-AUC differences against reconstructive baselines are statistically inconclusive.

Relative False-Alarm Suppression on Held-Out Telemetry: On the 19 non-GHL streams (Table II(a),(b)), latent predictive scoring via TS-JEPA and OpEntropy JEPA lowers held-out nominal FPR three- to four-fold under buffered attribution (9.3% and 10.1% versus 30.0% for TimesNet and 30.7% for TranAD) and eight- to twenty-fold under strictly causal streaming (2.4% and 2.8% versus 24.3% and 22.2%). Across all 33 streams, the paired buffered FPR reduction is statistically significant for both OpEntropy JEPA (−14.4% vs. TimesNet,

TABLE II
HELD-OUT CALIBRATION BENCHMARK (SPOT, $u{=}p_{98}$, $q{=}10^{-3}$): (A) RETROSPECTIVE BUFFERED ($S{=}64$) AND (B) STRICTLY CAUSAL TRAILING ATTRIBUTION ACROSS 19 NON-GHL STREAMS (95% STREAM-BOOTSTRAP CIS IN BRACKETS) AND 14 GHL STREAMS, (C) PER-DATASET BREAKDOWN (MEAN ± SD), AND (D) 130-STREAM SINGLE-SEED CROSS-CHECK. BEST IN **BOLD**, SECOND UNDERLINED.

	(a) Buffered, non-GHL (19 streams)					(b) Strictly causal, non-GHL (19 streams)				GHL buf (14)		Summary
Model	FPR [95% CI] ↓	F1 [95% CI] ↑	PR-AUC ↑	ER ↑	Δt ↓	FPR [95% CI] ↓	F1 [95% CI] ↑	ER ↑	Δt ↓	FPR ↓	F1 ↑	DS-Macro F1 ↑
Latent-space predictive architectures (JEPAs)												
TS-JEPA (unconstrained)	0.093 [.048, .145]	0.206 [.113, .319]	0.331	0.477	110.3	0.024 [.007, .046]	0.161 [.063, .287]	0.487	124.0	0.004	0.000	0.115
Operator-Entropy JEPA	0.101 [.043, .180]	**0.233** [.131, .355]	**0.351**	0.550	79.5	0.028 [.007, .063]	0.153 [.070, .247]	0.517	112.5	0.012	0.001	**0.125**
Reynolds-Stress JEPA	0.074 [.038, .118]	0.190 [.101, .295]	0.313	0.457	160.9	0.011 [.006, .017]	0.131 [.051, .232]	0.491	142.1	**0.003**	0.002	0.101
Potential-Flow JEPA	0.087 [.041, .142]	0.138 [.062, .231]	0.235	0.449	104.1	**0.008** [.002, .016]	0.029 [.005, .064]	0.459	138.4	0.011	0.006	0.089
Observation-space reconstructive baselines												
TimesNet	0.300 [.162, .446]	0.197 [.098, .315]	0.329	**0.583**	**48.5**	0.243 [.123, .375]	**0.209** [.109, .326]	**0.588**	**62.2**	0.061	**0.022**	0.100
TranAD	0.307 [.177, .445]	0.130 [.053, .232]	0.287	0.490	234.4	0.222 [.119, .337]	0.140 [.058, .241]	0.533	241.0	0.046	0.020	0.069
Non-neural linear baseline												
Linear AR Forecaster	**0.005** [.001, .011]	0.051 [.012, .104]	0.255	0.432	87.8	–	–	–	–	**0.002**	0.000	0.038

	(c) Dataset descriptors				Buffered Point-F1 (mean ± SD)				Buffered held-out FPR			
Dataset	n	K	π	$F1_{all\text{-}pos}$	TS-JEPA	OpEnt	TimesNet	TranAD	TS-JEPA	OpEnt	TimesNet	TranAD
Daphnet	6	9	.148	.256	0.145±0.058	0.122±0.061	0.229±0.232	0.144±0.119	0.091	0.059	0.313	0.298
GHL	14	16	.013	.026	0.000±0.000	0.001±0.003	0.022±0.030	0.020±0.025	0.004	0.012	0.061	0.046
Genesis	1	18	.003	.006	0.000±0.000	0.008±0.017	0.000±0.000	0.000±0.000	0.017	0.082	0.000	0.000
MSL	4	55	.255	.381	0.161±0.119	0.323±0.263	0.158±0.146	0.151±0.149	0.095	0.177	0.133	0.241
SMAP	6	25	.150	.236	0.379±0.361	0.396±0.329	0.289±0.334	0.167±0.341	0.135	0.125	0.549	0.513
Metro	1	5	.001	.003	0.000±0.000	0.000±0.000	0.001±0.001	0.000±0.000	0.014	0.026	0.000	0.000
SWAN	1	38	.279	.437	0.119±0.048	0.024±0.031	0.000±0.000	0.000±0.000	0.000	0.000	0.000	0.000

(d) 130-stream cross-check (Point-F1, dataset-macro / stream mean): NCAD 0.058/0.088; DCdetector 0.070/0.097; Anomaly Transformer 0.091/0.082; TimesNet 0.129/0.151; TranAD 0.156/0.152; TS-JEPA 0.160/0.186.

95% CI $[-24.1\%, -6.5\%]$, $p = 0.006$; -14.2% vs. TranAD, $p = 0.006$) and TS-JEPA (-15.3%, $p = 0.002$; -15.1%, $p = 0.003$). Dataset-clustered permutation tests across the $N = 7$ domain families remain significant ($p = 0.016$–0.031), and leave-one-dataset-out (LODO) folds hold macro buffered FPR to 3.8%–7.0% for JEPAs versus 8.4%–18.3% for reconstructors. On MSL, TimesNet achieves a lower buffered FPR than OpEntropy (13.3% vs. 17.7%), while on Genesis, Metro, and SWAN reconstructors emit zero alarms and achieve zero F1 due to overly high calibration thresholds.

Point-F1, Event Recall, and Ranking Parity: Improved false-alarm control does not translate into universal detection superiority. Across the 19 non-GHL streams in Table II(a), buffered Point-F1 and event recall for TS-JEPA (0.206, recall 0.477) and OpEntropy JEPA (0.233, recall 0.550) are comparable to TimesNet (0.197, recall 0.583) and TranAD (0.130, recall 0.490). Across all 33 streams, paired differences against TimesNet ($+0.012$ Point-F1, $p = 0.54$; -0.000 PR-AUC, $p = 0.39$) and TranAD ($+0.051$ Point-F1, $p = 0.88$; $+0.023$ PR-AUC, $p = 0.16$) are statistically indistinguishable from zero. As shown in Table II(c), between-stream standard deviations within each dataset are comparable to the means; only SMAP exceeds the trivial constant-alarm baseline $F1_{all\text{-}pos} = 0.236$ (0.379 and 0.396), although threshold-free PR-AUC exceeds base prevalence π on both SMAP (0.504–0.525 vs. $\pi = 0.150$) and MSL (0.422–0.434 vs. $\pi = 0.255$).

Domain and Operational Applicability Boundaries: Synthesizing performance across fault classes and sampling rates defines four operational regimes:

1) *Favorable operating regime:* Multi-channel spacecraft and wearable kinematic arrays ($K \geq 9$, SMAP, MSL, Daphnet) with persistent localized transients, where suppressing nuisance false alarms under automated calibration is the primary operational requirement.
2) *Latency-constrained regime:* Low-rate telemetry (0.1 Hz spacecraft or hourly Metro traffic) where a 63-step score-emission lag at $S = 64$ (10.5 min to 63 h) is prohibitive for immediate tripwires and requires short horizons $S \in \{1, 4\}$ (Section VII-D).
3) *Quiet single-channel faults:* Flatlines, stuck sensors, and dropouts (0.02–0.07 point recall, Table I(d)), which LayerNorm embeddings absorb as nominal background and which require auxiliary variance monitors at higher alarm cost (14%–15% FPR).
4) *Continuous non-stationary process ramps:* Industrial heating loops and traffic drifts (GHL, Metro; 0.000–0.022 F1), which defeat both static and rolling thresholds.

B. Evaluating Physical Inductive Priors: Conditioning vs. Detection

What does the Operator-Entropy formulation contribute beyond unconstrained TS-JEPA? Isolating the spectral regularizer on the fixed Koopman backbone ($\lambda_{\text{ent}} = 0$ control, Table I(a)) confirms strict TOST equivalence down to $\Delta = \pm 0.015$ and ± 0.010 Point-F1 (Section VI-A). However, the OpEntropy transition architecture (combining the linear Koopman operator $\mathbf{K}$, detached Lyapunov covariance alignment, and pre-LayerNorm relative radial energy scoring) shortens mean buffered detection delay from 110.3 steps in TS-JEPA to 79.5 steps (and from 124.0 to 112.5 steps causally in Table II(b), or $127.0 \rightarrow 58.2$ steps on the eight-stream decoupling panel). Cross-architecture Point-F1 differences against TS-JEPA satisfy TOST(± 0.05) across all 33 streams for both OpEntropy ($+0.016$, $p_{\text{TOST}} = 0.0019$) and Reynolds-Stress (-0.008, $p_{\text{TOST}} = 0.0017$), whereas Potential-Flow degrades Point-F1 (-0.081, $p = 0.033$ unadjusted) because its curl-free gradient parameterization penalizes periodic nominal orbits. The spectral-entropy and Reynolds priors therefore act as well-posed conditioning constraints rather than detection drivers.

C. Operational Incident Evaluation and Causal-vs-Buffered Trade-offs

Strictly causal trailing attribution represents real-time streaming deployment, whereas buffered attribution represents retrospective fixed-lag analysis. Comparing Panels (a) and (b) of Table II highlights a clear operational trade-off. Eliminating the 64-step lookahead buffer lowers nominal false alarms

from 9.3%–10.1% to 2.4%–2.8% for TS-JEPA and OpEntropy JEPA (and 1.1% for Reynolds-Stress), compared to 22.2%–24.3% for TranAD and TimesNet. Without retrospective smoothing, causal Point-F1 (0.153–0.161 vs. 0.209 for TimesNet) and causal event recall (0.487–0.517 vs. 0.588) are more conservative, and mean onset-to-alarm delay increases (Δt = 112.5–124.0 steps vs. 62.2 steps for TimesNet). In online streaming without lookahead buffers, latent predictive scoring thus functions as a conservative high-precision filter that trades raw onset speed for an 8–20× reduction in false alarms. On five non-stationary incident streams, pairing latent scores with dynamic rolling SPOT suppresses false alarms by up to 21× under regime drift.

D. Horizon Length, Score-Emission Latency, and False-Alarm Control

At target horizon $S = 64$, window scores emit $S - 1 = 63$ steps after window opening, which is distinct from empirical fault-detection delay $\Delta t = t_{\text{alarm}} - t^*$ measured after anomaly onset (Fig. 2, Row 3). To evaluate whether shorter horizons eliminate emission latency without sacrificing false-alarm control, Table III(a) retrains models across $S \in \{1, 4, 16, 32, 64\}$ on five streams (three seeds). At $S = 1$ (zero emission lag), OpEntropy JEPA holds causal FPR to 0.5% (vs. 2.5% for TimesNet) while achieving 0.729 event recall and a detection delay of $\Delta t = 76.3$ steps. As horizon S increases from 1 to 64, causal FPR for TimesNet rises nearly nine-fold (2.5% → 21.3%), whereas OpEntropy and TS-JEPA remain bounded within 0.3%–2.0%. Time-critical monitoring applications can therefore operate at $S \in \{1, 4\}$ to eliminate score-emission latency while preserving relative false-alarm suppression.

E. Deployable Calibration Quantiles vs. Matched Realized Test-FPR Operating Points

Because reconstructive baselines calibrate to a much higher realized false-alarm rate (22%–31%) than latent predictors (1%–10%), raw recall at a single SPOT threshold confounds discriminative sensitivity with threshold placement. Indeed, when anomaly segments span hundreds to over a thousand timesteps (mean length 402 steps in Daphnet and 1,102 steps in SMAP), a detector alarming randomly at 30% FPR intersects > 95% of segments by chance alone ($1 - (1 - \text{FPR})^L$). We therefore evaluate two complementary operating regimes:

1) **Deployable nominal calibration quantiles (q^*, Table III(b)):** Thresholds are set at fixed empirical quantiles of the nominal calibration block without test labels. At causal calibration quantile $q^* = 0.050$, TS-JEPA and OpEntropy JEPA achieve higher event recall (0.730 and 0.752) and higher precision (0.251 and 0.259) at fewer false alarms (177–203 per 1,000 nominal steps) than TimesNet (0.618 recall, 262 alarms) and TranAD (0.602 recall, 270 alarms).
2) **Post-hoc matched realized test-FPR curves (f^*, Fig. 3(b) and Table I(e)):** Oracle thresholds are chosen to equalize realized test nominal FPR across models. At a stringent 1% realized causal false-alarm budget ($f^* = 0.01$), latent models nearly double event recall (0.371–0.431) compared to TimesNet (0.217) and TranAD (0.149). At relaxed budgets ($f^* \geq 0.10$ causal and across buffered mappings), recall curves converge ($p = 0.625$ over $f^* \in [0.10, 0.50]$). Latent scoring thus specifically improves sensitivity in the low-false-alarm operating regime ($f^* \leq 0.01$).

F. Automated Threshold Calibration Realities on Non-Stationary Telemetry

Does static extreme-value thresholding achieve its nominal design risk $q = 10^{-3}$ (0.1%) out of sample? It does not. Across all 33 streams (Table IV(a)), standard SPOT achieves stream-macro (cell-weighted) causal FPR of 1.51% (1.71%) for TS-JEPA and 1.95% (2.12%) for OpEntropy, which is roughly 15–20× the 0.1% design target, and buffered stream-macro (cell-weighted) FPR of 5.54% (6.40%) and 6.38% (7.25%), or 55–64× (64–73×) the design target. Although this is an order-of-magnitude improvement over TimesNet (16.11% causal, 19.85% buffered) and TranAD (14.23% causal, 19.63% buffered; 142–222× overshoot), a 1.5%–6.4% nuisance-alarm rate still falls short of strict 0.1% operational targets. Closing this remaining 15–64× calibration gap requires pairing latent residuals with (i) block-declustered GPD fitting (replacing raw exceedances N_u with cluster maxima scaled by the Ferro–Segers extremal index $\hat{\theta}$), (ii) online conformal or rolling quantile updates gated by prediction-error variance under discrete regime shifts (Table IV(b)), or (iii) state-conditioned threshold tables indexed by telemetry operating mode. Three diagnostics in Table IV explain the static overshoot:

1) **GPD tail stability and quantile/length sensitivity:** On the 19 non-GHL streams at $u = p_{98}$, median calibration exceedance count is $N_u = 28$ causally (21 buffered) with 0% numerical solver fallback and low bootstrap threshold uncertainty ($\text{SE}(\tau)/\tau$ = 2.6%–3.5% causal). Causally, Grimshaw GPD fits hit the lower shape boundary ($\xi = -0.90$) in only 4.4% of JEPA cells (median $\xi = -0.27$) versus 34.4% for TimesNet and 54.4% for TranAD. Varying $u \in \{p_{95}, p_{98}, p_{99}\}$ leaves causal FPR unchanged (2.41%–2.42% for TS-JEPA), whereas halving calibration length (50% N_{cal}) raises causal FPR to 3.72% (TS-JEPA) and 4.13% (OpEntropy).
2) **Platykurtic collapse under total-threshold kurtosis scaling:** While total-threshold kurtosis scaling $\tau_{\text{eff}} = \tau[1 + \frac{1}{6}\tanh(\gamma_2)]$ reduces buffered FPR on leptokurtic cells ($\gamma_2 \geq 0$, 4.49% → 1.11% for TS-JEPA), 64-step buffered smoothing renders 31.9% of TS-JEPA cells platykurtic ($\gamma_2 < 0$) with tight relative dispersion ($\sigma_{\text{cal}}/\mu_{\text{cal}} \approx$ 0.02–0.05). Scaling total τ toward zero pulls τ_{eff} below the $u = p_{98}$ floor in 28.4% of cells, inflating overall buffered FPR from 6.40% to 16.30% (Table IV(a)). Scaling strictly the tail excess above u, $\tau_{\text{exc}} = u + (\tau - u)[1 + \frac{1}{6}\tanh(\gamma_2)] \geq u$, prevents threshold collapse below u and holds 19-stream buffered FPR to 9.19% vs. 9.29% unadjusted (2.40% vs. 2.42% causal).
3) **Three-stage refitted permutation decomposition (Table IV(c)):** Simply permuting held-out evaluation scores against a fixed threshold cannot change the exceedance rate. Instead, our three-stage protocol pools the nominal calibra-

TABLE III
OPERATIONAL LATENCY AND CALIBRATION-QUANTILE TRADE-OFFS: (A) TARGET-HORIZON SWEEP ($S \in \{1, 4, 16, 32, 64\}$, LAG $S-1$, n=5 STREAMS, 3 SEEDS) AND (B) DETECTION AT DEPLOYABLE NOMINAL CALIBRATION QUANTILES q^* (n=19 STREAMS; ER = EVENT RECALL, P = PRECISION, FA = FALSE ALARMS PER 1,000 STEPS).

(a) Target-Horizon and Score-Emission Sweep

		OpEntropy, causal			TS-JEPA, causal		TimesNet, causal		OpEntropy, buf	
S	Lag	FPR	ER	Δt	FPR	ER	FPR	ER	FPR	ER
1	0	0.005	0.729	76.3	0.004	0.524	0.025	0.895	0.005	0.729
4	3	0.007	0.624	179.3	0.006	0.395	0.031	0.910	0.020	0.481
16	15	0.003	0.257	200.3	0.003	0.257	0.061	0.700	0.043	0.391
32	31	0.011	0.219	397.9	0.003	0.276	0.098	0.657	0.057	0.376
64	63	0.020	0.410	209.8	0.004	0.200	0.213	0.614	0.099	0.367

(b) Detection at Matched Nominal Budgets q^*

		TS-JEPA			OpEntropy			TimesNet			TranAD		
Regime	q^*	ER	P	FA	ER	P	FA	ER	P	FA	ER	P	FA
Causal	0.001	0.493	0.285	88.1	0.556	0.228	93.7	0.590	0.219	222.4	0.533	0.156	209.6
Causal	0.050	0.730	0.251	177.3	0.752	0.259	202.8	0.618	0.221	262.3	0.602	0.159	269.5
Buffered	0.001	0.487	0.236	219.4	0.559	0.242	247.3	0.583	0.194	259.2	0.516	0.130	262.6
Buffered	0.050	0.582	0.233	283.7	0.654	0.268	317.6	0.612	0.196	287.3	0.596	0.145	295.3

TABLE IV
THRESHOLD CALIBRATOR COMPARISON AT DESIGN RISK q=10^{-3}: (A) STATIC RULES ACROSS 33 STREAMS (CELL-WEIGHTED MEANS, STREAM-MACRO IN PARENTHESES), (B) CONTROLLED-DRIFT STRESS TEST (n=16), AND (C) THREE-STAGE REFITTED PERMUTATION DECOMPOSITION OF EXCESS CAUSAL FPR (n=18).

(a) Model	Calibrator Rule	Causal FPR	Overshoot $> q$	Buffered FPR	Point-F1 (buf)
TS-JEPA	Standard SPOT ($q = 10^{-3}$)	0.0171 (0.0151)	10%	0.0640 (0.0554)	0.1386
	SPOT + Kurtosis ($\beta = 0.167$)	0.0185 (0.0159)	10%	0.1630 (0.1397)	0.1498
	Empirical 99.5th Percentile	0.0254 (0.0229)	21%	0.0756 (0.0659)	0.1399
	Validation Maximum	0.0160 (0.0140)	7%	0.0658 (0.0568)	0.1385
OpEntropy	Standard SPOT ($q = 10^{-3}$)	0.0212 (0.0195)	14%	0.0725 (0.0638)	0.1573
	SPOT + Kurtosis ($\beta = 0.167$)	0.0209 (0.0188)	10%	0.1346 (0.1165)	0.1610
	Empirical 99.5th Percentile	0.0298 (0.0281)	27%	0.0882 (0.0783)	0.1633
	Validation Maximum	0.0187 (0.0170)	10%	0.0727 (0.0637)	0.1566
TimesNet	Standard SPOT ($q = 10^{-3}$)	0.1795 (0.1611)	49%	0.2217 (0.1985)	0.1395
	SPOT + Kurtosis ($\beta = 0.167$)	0.1843 (0.1623)	46%	0.2281 (0.2011)	0.1352
	Empirical 99.5th Percentile	0.1914 (0.1752)	51%	0.2338 (0.2125)	0.1413
	Validation Maximum	0.1791 (0.1599)	48%	0.2263 (0.2032)	0.1410
TranAD	Standard SPOT ($q = 10^{-3}$)	0.1601 (0.1423)	46%	0.2217 (0.1963)	0.0936
	SPOT + Kurtosis ($\beta = 0.167$)	0.1722 (0.1513)	43%	0.2453 (0.2151)	0.0940
	Empirical 99.5th Percentile	0.1712 (0.1544)	51%	0.2343 (0.2093)	0.0962
	Validation Maximum	0.1580 (0.1392)	44%	0.2270 (0.2009)	0.0952

(b) Injected condition	Static SPOT	Static empirical	Rolling empirical	Rolling SPOT
Stationary (no injection)	0.061	0.162	0.016	0.047
Mean shift $+1.0\sigma$	0.087	0.216	0.011	0.043
Regime change $+1\sigma$	0.073	0.187	0.015	0.070
Variance scale 2.0	0.082	0.195	0.016	0.047
Autocorrelation 0.9	0.113	0.232	0.024	0.055
Continuous mean ramp	0.075	0.186	0.326	0.385

	Refitted Stage Causal FPR			Estimated Share of Excess ($\text{FPR}_{\text{chrono}} - q$)		
(c) Model	$\text{FPR}_{\text{chrono}}$	$\text{FPR}_{\text{block}}$	$\text{FPR}_{\text{point}}$	Inter-Block Shift (95% CI)	Serial Dep. (95% CI)	i.i.d. Tail
TS-JEPA	0.0121	0.0058	0.0019	56.6% (+.0063 [−.0054, +.0194])	35.3% (+.0039 [+.0001, +.0103])	8.1%
OpEntropy	0.0185	0.0073	0.0022	64.1% (+.0112 [−.0015, +.0336])	29.1% (+.0051 [−.0005, +.0146])	6.9%
TimesNet	0.0872	0.0204	0.0037	77.5% (+.0668 [+.0310, +.1082])	19.4% (+.0167 [+.0041, +.0338])	3.1%
TranAD	0.0741	0.0067	0.0032	92.3% (+.0674 [+.0354, +.1062])	4.7% (+.0035 [+.0002, +.0078])	3.0%

tion and evaluation vectors $\mathbf{v}_{\text{pool}} = [v_{\text{cal}}, v_{\text{hold}}] \in \mathbb{R}^{N_{\text{cal}}+N_{\text{hold}}}$ and *refits* the SPOT threshold ($u = p_{98}, q = 10^{-3}$) from scratch on each permuted calibration split: Stage 1 (chronological) fits τ_{chrono} on v_{cal} and evaluates $\text{FPR}_{\text{chrono}}$ on v_{hold}; Stage 2 ($L = 64$ block permutation) shuffles contiguous 64-step blocks across $\mathbf{v}_{\text{pool}}$, splits into permuted calibration (N_{cal}) and evaluation (N_{hold}) halves, *refits* τ_{block}, and evaluates $\text{FPR}_{\text{block}}$ (equalizing macro-block distributions while preserving intra-block autocorrelation); and Stage 3 (pointwise i.i.d. permutation) uniformly shuffles all points in $\mathbf{v}_{\text{pool}}$, splits into calibration and evaluation halves, *refits* τ_{point}, and evaluates $\text{FPR}_{\text{point}}$. Across the 18 streams with $N_{\text{cal}}, N_{\text{hold}} \geq 100$ (Table IV(c)), inter-block distribution shift ($\text{FPR}_{\text{chrono}} - \text{FPR}_{\text{block}}$) accounts for an estimated 56.6%–64.1% of excess causal overshoot for JEPAs (77.5%–92.3% for reconstructors), and serial dependence ($\text{FPR}_{\text{block}} - \text{FPR}_{\text{point}}$) for 29.1%–35.3%. Because latent predictors achieve $\text{FPR}_{\text{chrono}} = 0$ on 11 of 18 streams (concentrating stream-level variance onto the seven drifting streams so the TS-JEPA and OpEntropy inter-block 95% CIs in Table IV(c) span zero), these percentages represent estimated shares of aggregate excess risk rather than invariant per-stream causal proportions.

VIII. CONCLUSION AND FUTURE WORK

We evaluated how Joint-Embedding Predictive Architectures condition anomaly residuals under automated extreme-value calibration across 33 continuous telemetry streams under a chronological held-out protocol:

1) **Mechanism Isolation and Regularizer Equivalence:** Comparing 12 controlled backbones decouples score compression from representation learning: LayerNorm bottlenecks bound residual kurtosis and halve false alarms, whereas learned latent projections halve calibration-to-holdout distribution drift ($D_{\text{KS}} : 0.376 \rightarrow 0.209$–$0.214$) and reduce causal false-positive rates to 0.9%–2.8%. On a fixed Koopman backbone, the spectral-entropy regularizer satisfies TOST equivalence down to $\Delta = \pm 0.010$ Point-F1, functioning as an operator conditioning constraint while the OpEntropy transition architecture shortens detection delay relative to unconstrained TS-JEPA ($110.3 \rightarrow 79.5$ steps buffered; $127.0 \rightarrow 58.2$ causal).
2) **Relative False-Alarm Control vs. Detection Parity:** Latent predictive scoring lowers held-out false alarms three- to four-fold under buffered attribution and eight- to twenty-fold under strictly causal streaming, and nearly doubles causal event recall at stringent realized false-alarm budgets ($f^* = 0.01$), while overall Point-F1 and PR-AUC remain comparable to observation-space reconstructors.
3) **Calibration Drift and Operational Boundaries:** Static SPOT still overshoots its nominal 0.1% design risk by 15–64× out of sample, driven primarily by inter-block distribution shift (56.6%–64.1% of excess risk) and serial dependence (29.1%–35.3%). Future work should pair latent predictive residuals with declustered GPD fitting, online conformal quantile updates, and hybrid variance monitors to close the remaining calibration gap and cover quiet single-channel flatlines.

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